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 - Orientation: Portrait (except for special diagrams)
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- c. Margins:
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 - Body Content: 11
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 - Top Right (defined in the template)
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**Development of an Interactive Learning Management System
for Computer System Servicing (CSS) Students at
Paliparan III Senior High School**

AN UNDERGRADUATE CAPSTONE PROPOSAL

**Presented to the Faculty of the
Institute of Computing and Digital Innovations
Kolehiyo ng Lungsod ng Dasmariñas
City of Dasmariñas, Cavite, Philippines**

**In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Systems**

Diane Mariel C. Barcelon

Jaira Kae V. Bono

Ben Joseph R. Pinpin

Anthony C. Ponce

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Introduction

In today's digital age, education continues to evolve through the integration of technology in the teaching and learning process. Schools are increasingly adopting digital platforms to enhance accessibility to learning materials, streamline academic processes and monitor student performance efficiently. Learning Management Systems (LMS) play a vital role in modern education by providing centralized platforms for organizing lessons, assessment, and academic records.

At Paliparan III Senior High School, particularly among students who take specialization in Computer System Servicing (CSS) are expected to develop both theoretical knowledge and practical technical skills. As a technology focused field, CSS students' need structured learning materials, interactive activities, and consistent monitoring of academic progress for developing their competencies. However, the school currently lacks a centralized digital platform these needs.

Currently learning materials distributed through presentations, printed modules, and files sent through messengers while teachers conduct their academic evaluations through manual methods. This situation limits student's ability to access previous lessons needed for review or advanced study. Teachers also encounter difficulties in tracking student performance while providing feedback to students within limited time frames. The lack of a centralized digital platform affects both teaching methods and student participation in learning activities.



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According to Duch, May, & George (2024) the use of Learning Management Systems (LMS) has grown in popularity primarily for their ability to manage and organize digital information related to teaching and learning. To address the growing need for a more efficient and accessible learning environment, this study proposes the development of an interactive learning management system designed specifically for CSS students of Paliparan III Senior High School.

The system aims to provide an interactive online platform that enhances both teaching and students learning by integrating learning management functionalities such as class management, announcements, assignments, and grading with interactive learning modules that can support students self-paced learning. These modules include videos, flashcards, quizzes to enhance students understanding and engagement. The system also has automated progress tracking tool which evaluates student performance based on quiz results and engagement levels. This tool will generate visual reports and dashboards for students, promoting transparency and a deeper understanding of academic progress. While teachers can view summarized student progress for monitoring purposes, they do not directly control or modify module results, ensuring a clear distinction between the systems tracking students performance automatically and teacher managed assessments.

By combining these core functions, the Interactive Learning Management System seeks to improve academic monitoring, streamline communication between



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teachers and learners and foster a more interactive and data-driven educational environment. Furthermore, the system is designed with adaptability in mind, enabling the integration of additional subjects and modules, which allows it to remain relevant and applicable even as academic structures and curriculum requirements evolve. This project supports the school's commitment to embracing innovation and enhancing the quality of education through technology.

Project Context.

The Computer System Servicing (CSS) students at Paliparan III Senior High School currently rely on traditional teaching methods which include classroom instruction with presentations, printed modules and use manual methods. While these methods provide fundamental teaching support, the school lacks a digital system that integrates all learning materials, assessment tools and student performance information into a single platform.

For students, the absence of such a platform significantly limits their learning opportunities. Students do not have access to a system where they can review past lessons, study upcoming topics, or monitor their academic progress in real time. Learning materials are difficult to retrieve as they are scattered across various formats and platforms. As a result, students have limited opportunities to engage in self-directed and advanced study beyond what is covered in class.



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From the teachers' perspective, the lack of a centralized platform makes it difficult to efficiently monitor and evaluate student performance. Tasks such as tracking quiz results, checking module completion, and preparing performance reports require an additional time and effort when done manually. This limits the teachers' ability to provide timely feedback and data-driven interventions for students who may need academic support.

These challenges reveal a clear gap in the current academic process for CSS students at Paliparan III Senior High School. The lack of a centralized digital platform for managing learning materials, assessments, and student performance results in inefficiencies in both teaching and learning. This situation highlights the need for a more organized and accessible system to support academic activities.

Statement of the Problem

The purpose of this study is to develop an Interactive Learning Management System designed to support CSS (Computer Systems Servicing) students of Paliparan III Senior High School in improving learning efficiency, engagement, and performance monitoring.

Specifically, it seeks to address the following problems:



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1. How can the system provide an organized and accessible digital platform where CSS students can easily access learning materials, modules, and assessments to improve their learning experience?
2. How can the system provide interactive learning tools such as videos, flashcards and quizzes that help students understand CSS concepts more effectively?
3. How can the system automate the tracking of student performance based on assessments, module completion, and quizzes?
4. How can the system generate automated progress reports that help both students and teachers identify strengths, weaknesses, and areas needing improvement?
5. How can the system support flexible module management to accommodate changes in curriculum and subject offerings?

Objectives of the Study

The primary objective of this study is to develop an Interactive Learning Management System that provides an efficient online platform for CSS students and teachers at Paliparan III Senior Highschool to manage lessons, quizzes, and performance monitoring through automated progress tracking.

Specific Objectives:



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This study aims to:

1. Design and develop an Interactive Learning Management System for CSS students at Paliparan III Senior High School that will be capable of:
 - a. A user-friendly web-based interface for both teachers and students to easily use the system.
 - b. User Authentication and Role Management to ensures secure access and proper identification of users before entering the platform.
 - c. Super Admin Management has the full control over the system including managing students, teacher and admin accounts, viewing all users, monitor system activity and approving or rejecting modules created by teachers before they are published and made available to students in the learning module section of the system.
 - d. Admin Management to allow administrators to manage teacher and student accounts, create subjects and sections, assign subjects and sections to teachers, and automatically generate class codes upon assignment, as well as view student information and their assigned teachers.



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e. Teacher Dashboard to provide teachers with tools for uploading lessons, managing quizzes and grades, posting announcements, and managing student performance efficiently.

f. Student Dashboard provides students with all learning resources and academic progress in one place. This also allow students to view enrolled subjects, access learning materials, and take quizzes or submit assignments.

g. Assignment Dashboard allow teachers to manage student submissions, and manually record student grades through the system.

h. Interactive learning modules provides pre-built modules for Grade 11 and Grade 12 CSS subjects that include multimedia elements such as videos, flashcards, quizzes, and activities, while also allowing teachers to create and integrate additional modules to support curriculum changes.

i. Automated Progress Tracking – tracks student performance based on module completion, and quiz results, generating visual progress reports that students can view to monitor their own learning.

j. Announcements and Notifications to keep both teachers and students informed about important updates, deadlines, and activities through timely notifications and announcements.



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k. Reports and Analytics generate detailed performance reports on student performance based on submitted assignments and assessments, with filters such as subject, grade level, and section.

l. Announcements and Notifications to keep both teachers and students informed about important updates, deadlines, and activities through timely notifications within the platform.

2. Identify the software and hardware requirements necessary for the development and deployment of the system.

3. Test the performance of the system through unit, integration, and usability testing involving:

a. Teachers

b. IT professionals

c. Non-technical users

4. Evaluate the system's quality based on the ISO 25010 Software Quality Standards, focusing on functionality, reliability, usability, and efficiency.



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Conceptual Framework

Figure 1

Interactive Learning Management System for CSS Students

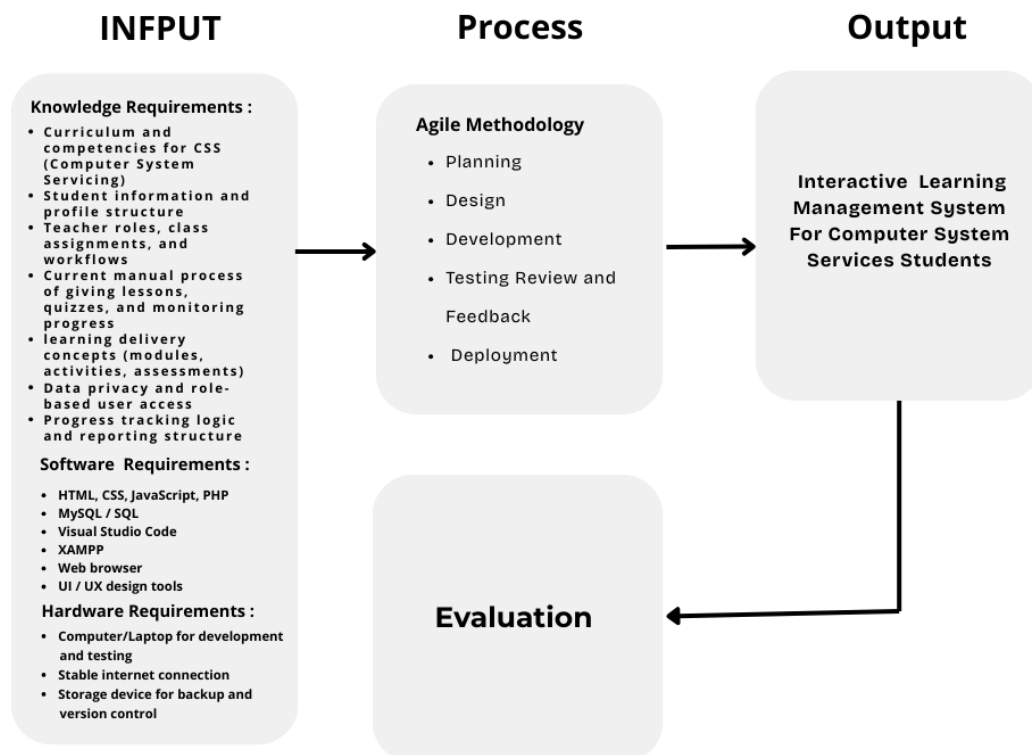


Figure 1: The Input phase is based on three essential resource categories: Knowledge Requirements, Software Resources, and Hardware Resources. These combined inputs provide the foundation for the system's architecture and development. It begins with Knowledge Requirements. This phase is where the indispensable data and operational specifications: this means acquiring the complete



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official CSS curriculum and competencies to make sure the system's content is accurate. It also means writing down the current school practices, like the existing manual method of teaching lessons and recording progress, which the new system will either take over or assimilate. While, the progress tracking logic and data privacy/user access rules are the key factors to making the system workable and safe at the very beginning. The Software Requirements make it clear what the development stack is, for example, PHP for server-side programming, MySQL/SQL for the powerful database that will store the student and curriculum data, and client-side technologies like HTML, CSS, and JavaScript for the interactive part of the website. The Hardware Requirements, in the end, make sure that there are enough physical resources available, such as the required computing devices and a reliable internet connection for the uninterrupted development and ultimately, testing.

Figure 1: The process is based on the Agile Methodology, which is characterized by different iterations throughout its stages. The planning phase consists of taking the requirements from the Input phase and determining the features and user stories for the current iteration (or sprint). The design phase converts these requirements into technical specifications, which also includes making the entity-relationship diagrams for the MySQL database and crafting the interface (UI) and experience (UX) to guarantee that the system is user-friendly for both students and teachers. The development phase is concerned with coding, and within this phase, the team



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develops the software with the selected technologies. Testing Review and Feedback returning to the design or development phases. The last stage is Deployment, which encompasses putting the Interactive learning management final and stable version on the servers so that it can be used by the students and teachers of Paliparan III Senior High School. Back is a cyclical and crucial procedure: the developers and the stakeholders verify the system's functionality (for instance, checking that the automated progress tracking formula is accurate) and give their input to sharpen features, which frequently results in returning to the design or development phases. The last stage is Deployment, which encompasses putting the Interactive E-LMS's final and stable version on the servers so that it can be used by the students and teachers of Paliparan III Senior High School.

Figure 1: The Output of this study is the complete, and fully functional system of Interactive Learning Management System for CSS students that directly results from the successful execution of the Process phase. This is the main deliverable that brings together all the essential knowledge and technical requirements stipulated. The system is interactive providing lively digital modules, quizzes, and assessments that are in line with the CSS curriculum. The most important feature is the Automated Progress Tracking, which assesses, logs, and gives feedback on the students' performance instantly, thus doing away with the manual record-keeping process. The



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system is implemented and functional, it has become the new platform for the transfer of knowledge and monitoring of performance for the intended students.

Figure 1: The evaluation of will be guided by the ISO/IEC 25010 Software Quality Standard to ensure the system meets essential criteria for functionality and usability. The system will undergo a series of assessments after its development and initial deployment. It will be evaluated according to the various ISO 25010 quality characteristics including Functional Suitability, where the existence of different features, such as automated progress tracking, and interactive module, will be verified. Performance Efficiency will determine how fast and to what extent the system processes data and produces outputs, particularly when multiple branches use the system simultaneously. The Usability will be assessed through surveys and feedback from students and teachers regarding interface comprehension, navigability, and overall user experience. The system's Reliability will be observed in realistic scenarios where its stability will be evaluated over a certain period and also monitoring errorfree or downtime data input consistency. Security is another important ISO parameter that will evaluate the safeguarding of sensitive data related to students and tutors with an emphasis on data confidentiality, access rights, and protection of records at the branch level. The system's performance in terms of Maintainability and Portability will be tested so as to certify that it can be updated easily, debugged, and customized for the system. This systematic assessment, utilizing the ISO 25010 framework, will enable



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the conclusive validation of the systems effectiveness, quality, and conformity to the operational needs.

Scope and Limitations

Scope

This study focuses on the development of an Interactive Learning Management System designed primarily for Computer System Servicing (CSS) students and teachers of Paliparan III Senior High School. The system provides a web-based platform that integrates learning management functionalities such as class management, announcements, assignments, and grading, along with interactive learning modules that support self-paced learning.

The system allows students to access learning materials, participate in interactive modules, complete quizzes and activities, submit assignments, and monitor their own progress through system-generated tracking tools. Teachers can manage classes, upload learning materials, create and evaluate assignments, post announcements, and monitor student performance through reports. Additionally, teachers are allowed to create and integrate their own learning modules, which are subject to approval by the Super Admin before being published in the system.



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The system includes four types of users: Super Admin, Admin, Teacher, and Student. The Super Admin has full control over the system, including managing all user accounts, monitoring system activity, and approving teacher-created modules. The Admin manages student and teacher accounts, subjects, and class sections. Teachers handle instructional and assessment-related tasks, while students serve as the primary users of the learning platform.

Furthermore, the system is designed to support adaptability by allowing the integration of additional modules and subjects, making it capable of accommodating possible curriculum changes such as the transition from strand-based to elective-based learning.

Limitations

This study is limited to the development and implementation of the system within the context of Paliparan III Senior High School and is primarily focused on CSS-related subjects. Although the system supports the addition of other modules, it does not fully implement a complete curriculum management system for all academic strands.

The interactive learning modules are pre-built and system-driven, and while teachers can create additional modules, they cannot modify the automated progress tracking



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results generated by the system. The grading functionality is limited to assignments and assessments created by teachers within the LMS.

The system does not include advanced features such as artificial intelligence-based recommendations, real-time collaboration tools, or offline access. Additionally, the evaluation of the system is limited to functionality, usability, reliability, and efficiency based on ISO 25010 standards, and does not cover long-term deployment or large-scale implementation.

Definition of Terms

Administrator – In this study, the administrator refers to the authorized user responsible for managing teacher accounts, assigning subjects and sections, and overseeing system data within the Interactive Learning Management System.

Assessment – Refers to the quizzes and long examinations integrated into the system to evaluate students' understanding of CSS lessons.

Computer System Servicing (CSS) – Refers to the senior high school technical-vocational strand focused on computer hardware servicing and related competencies, which serves as the target users of the developed system.



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Dashboard – Refers to the user interface within the system that displays relevant information such as enrolled subjects, uploaded modules, quiz results, and progress reports for students and teachers.

Grade Computation – Refers to the automated calculation of quiz and assessment scores performed by the system.

Interactive Learning Management System (ILMS) – In this study, it refers to the web-based platform developed to deliver interactive learning modules, manage assessments, and monitor the academic performance of CSS students.

Interactive Learning Module – Refers to the digital instructional content within the system that includes lessons, videos, flashcards, quizzes, and activities designed to enhance student engagement.

Progress Tracking – Refers to the system feature that monitors and displays student completion of modules, quiz results, and overall academic performance.

Usability Testing – Refers to the evaluation process conducted with selected users to determine the system's ease of use, functionality, and overall user experience.

Web-Based System – Refers to the online platform accessible through internet browsers without requiring installation of a separate application.



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REFERENCES

Duch, D., May, M., & George, S. (2024). Students' performance in learning management systems: An approach to key attributes identification and predictive algorithm design. In *Proceedings of the 18th International Conference on Computer Supported Education (CSEDU 2024)* (pp. 127–142). SCITEPRESS.

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